

COACHING PROGRAMMING QUESTIONS AND ANSWERS

1. SWAP 2 VARIABLES WITH 3rd VARIABLE

```
def swap(a,b):  
    c = a  
    a = b  
    b = c  
    print(a,b)  
a = int(input("ENTER FIRST NUMBER : "))  
b = int(input("ENTER SECOND NUMBER : "))  
swap(a,b)
```

2. SWAP 2 VARIABLES WITHOUT USING 3rd VARIABLE

```
def swap(a,b):  
    a = a+b  
    b = a-b  
    a = a-b  
    print(a,b)  
a = int(input("ENTER FIRST NUMBER : "))  
b = int(input("ENTER ANOTHER NUMBER : "))  
swap(a,b)
```

3. SWAP 2 VARIABLES USING PYTHON

```
def swap(a,b):  
    a,b = b,a  
    print(a,b)  
a = int(input("ENTER A NUMBER : "))  
b = int(input("ENTER ANOTHER NUMBER : "))  
swap(a,b)
```

4. SWAP 3 VARIABLES

```
def swap(a,b,c):  
    a = a+b+c  
    b = a-b-c  
    c = a-b-c  
    a = a-b-c  
    print(a,b,c)  
a = int(input("ENTER FIRST NUMBER : "))  
b = int(input("ENTER SECOND NUMBER : "))  
c = int(input("ENTER THIRD NUMBER : "))  
swap(a,b,c)
```

5. EVEN ODD

```
def evenodd(n):  
    if n%2 == 0:  
        return ("EVEN")  
    else:  
        return ("ODD")  
n = int(input("ENTER A NUMBER : "))  
print(evenodd(n))
```

6. POSITIVE NEGATIVE

```
def pos(n):  
    if n < 0:  
        print("NEGATIVE")  
    elif n > 0:  
        print("POSTIVE")  
    else:  
        print("ZERO")  
n = int(input("ENTER A NUMBER : "))  
pos(n)
```

7. LEAP YEAR

```
def leap(n):  
    if n%400 == 0:  
        print("LEAP YEAR")  
    else:  
        if n%4 == 0 and n%100 != 0:  
            print("LEAP YEAR")  
        else:  
            print("NOT A LEAP YEAR")  
n = int(input("ENTER A NUMBER : "))  
leap(n)
```

8. INT OR DECIMAL

```
def intee(n):  
    if n == int(n):  
        print("INTEGER")  
    else:  
        print("DECIMAL")  
a = eval(input("ENTER A NUMBER : "))  
intee(a)
```

9. SQRT

```
from math import sqrt
def sqrt_t(n):
    return sqrt(n)
a = int(input("ENTER A NUMBER : "))
print(sqrt_t(a))
```

10. POW

```
from math import pow
def power(n):
    print(pow(n,2))
n = int(input("ENTER A NUMBER : "))
power(n)
```

11. TABLE OF N

```
def table(n):
    for i in range(n,(n*10)+1,n):
        print(i)
n = int(input("ENTER A NUMBER : "))
table(n)
```

12. 1 TO N EVEN NUMBERS

```
def even(n):
    i = 2
    while(i <= n):
        print(i)
        i = i + 2
n = int(input("ENTER A NUMBER : "))
even(n)
```

13. 1 to N ODD

```
def odd(n):
    i = 1
    while(i <= n):
        print(i)
        i = i + 2
```

```
n = int(input("ENTER A NUMBER : "))
odd(n)
```

14. SUM OF 1 TO N

```
def sum(n):
    i = 1
    sum = 0
    while(i <= n):
        sum = sum + i
        i = i + 1
    print(sum)
n = int(input("ENTER A NUMBER : "))
sum(n)
```

15. SUM OF SQUARE 1 TO N

```
def sumofsquares(n):
    i = 1
    sum = 0
    while(i <= n):
        sum = sum + (i **2)
        i = i + 1
    print(sum)
n = int(input("ENTER A NUMBER : "))
sumofsquares(n)
```

16. SUM OF CUBES FROM 1 TO N

```
def sumofcubes(n):
    i = 1
    sum = 0
    while(i <= n):
        sum = sum + (i **3)
        i = i + 1
    print(sum)
n = int(input("ENTER A NUMBER : "))
sumofcubes(n)
```

17. SUM OF FACTORS

```
def sumoffactors(n):  
    i = 1  
    sum = 0  
    while(i <= n//2):  
        if n%i == 0:  
            sum = sum + i  
        i = i + 1  
    print(sum)  
n = int(input("ENTER A NUMBER : "))  
sumoffactors(n)
```

18. COUNT DIGITS

```
def countdigits(n):  
    count = 0  
    while(n):  
        n = n//10  
        count = count + 1  
    print(count)  
n = int(input("ENTER A NUMBER : "))  
countdigits(n)
```

19. SUM OF DIGITS

```
def sumofdigits(n):  
    sum = 0  
    while(n):  
        b = n%10  
        sum = sum + b  
        n = n//10  
    print(sum)  
n = int(input("ENTER A NUMBER : "))  
sumofdigits(n)
```

20. REVERSE A NUMBER

```
def reverse(n):
    rev = 0
    while(n):
        b = n%10
        rev = (rev*10) + b
        n = n//10
    print(rev)
n = int(input("ENTER A NUMBER : "))
reverse(n)
```

21. PALINDROM

```
def palindrom(n):
    rev = 0
    a = n
    while(n):
        b = n%10
        rev = (rev * 10) + b
        n = n//10
    if rev == a:
        print("PALINDROM")
    else:
        print("NOT A PALINDROM")
n = int(input("ENTER A NUMBER : "))
palindrom(n)
```

22. SUM OF REVERSE AND NUMBER

```
def rev_sum(n):
    a = n
    rev = 0
    while(n):
        b = n%10
        rev = (rev * 10) + b
        n = n//10
    print(a + rev)
n = int(input("ENTER A NUMBER : "))
rev_sum(n)
```

23. ARMSTRONG

```
def countdigits(n):
    count = 0
    while(n):
        n = n//10
        count = count + 1
    return (count)
def arms(n):
    c = countdigits(n)
    a = n
    sum = 0
    while(n):
        b = n%10
        sum = sum + (b**c)
        n = n//10
    if sum == a:
        print("ARMSTRONG")
    else:
        print("NOT A ARMSTRONG")
n = int(input("ENTER A NUMBER : "))
arms(n)
```

24. DISARUM

```
def countdigits(n):
    count = 0
    while(n):
        n = n//10
        count = count + 1
    return (count)
def desa(n):
    c = countdigits(n)
    sum = 0
    a = n
    while(n):
        b = n%10
        sum = sum + (b**c)
        n = n//10
        c = c - 1
    if a == sum:
        print("DIASUM NUMBER")
    else:
        print("NOT A DIASUM NUMBER")
n = int(input("ENTER A NUMBER : "))
```

```
desa(n)
```

25. FACTORIAL

```
def factorail(n):  
    fact = 1  
    while(n):  
        fact = fact * n  
        n = n -1  
    print(fact)  
n = int(input("ENTER A NUMBER : "))  
factorail(n)
```

26. PRODUCT DIGITS

```
def prod(n):  
    prod = 1  
    while(n):  
        b = n%10  
        prod = prod * b  
        n = n//10  
    print(prod)  
n = int(input("ENTER A NUMBER : "))  
prod(n)
```

27. SPY NUMBER

```
def spy(n):  
    sum = 0  
    prod = 1  
    while(n):  
        b = n%10  
        sum = sum + b  
        prod = prod * b  
        n = n//10  
    if sum == prod:  
        print("SPY NUMBER")  
    else:  
        print("NOT A SPY NUMBER")  
n = int(input("ENTER A NUMBER"))  
spy(n)
```


28. PERFECT NUMBER

```
def perfect(n):
    i = 1
    sum = 0
    while(i <= n//2):
        if n%i == 0:
            sum = sum + i
        i = i + 1
    if sum == n:
        print("PERFECT NUMBER")
    else:
        print("NOT A PERFECT NUMBER")
n = int(input("ENTER A NUMBER : "))
perfect(n)
```

29. PRIME NUMBER

```
from math import sqrt
def prime(n):
    if n <= 1:
        return False
    i = 2
    end = int(sqrt(n))
    while(i <= end):
        if n%i == 0:
            return False
        i = i + 1
    return True
n = int(input("ENTER A NUMBER : "))
print(prime(n))
```

```
from math import sqrt
def prime(n):
    if n <= 1:
        return False
    end = int(sqrt(n))
    for i in range(2,end+1):
        if n%i == 0:
            return False
    return True
n = int(input("ENTER A NUMBER : "))
print(prime(n))
```

30. PRINTING FIRST N TERMS OF FIBONACCI SERIES

```
def fibo(n):
    if n == 1:
        print(0)
    elif n == 2:
        print(0)
        print(1)
    else:
        print(0)
        print(1)
        n = n - 2
        a = 0
        b = 1
        while(n):
            c = a+b
            print(c)
            a = b
            b = c
            n = n - 1
n = int(input("ENTER A NUMBER : "))
fibo(n)
```

31. FIND A NUMBER IF IT IS IN A FIBO SERIES

```
def find(n):
    if n == 0 or n == 1:
        return True
    a = 0
    b = 1
    while(1):
        c = a+b
        if c == n:
            return True
        if c > n:
            return False
        a = b
        b = c
n = int(input("ENTER A NUMBER : "))
find(n)
```

32. SUNNY

```
from math import sqrt
def sunny(n):
    n = n + 1
    x = sqrt(n)
    y = int(x)
    if x == y:
        return True
    return False
n = 15
sunny(n)
```

33. HAPPY NUMBER

```
def happy(n):
    sum = n
    while(sum > 9):
        local = 0
        while(sum != 0):
            b = sum%10
            local = local + (b*b)
            sum = sum//10
        sum = local
    if sum == 1 or sum == 7:
        return True
    return False
n = int(input())
print(happy(n))
```

34. SUM OF ARRAY

```
n = int(input("ENTER N : "))
arr = list(map(int,input().split()))
sum = 0
for i in range(n):
    sum = sum + arr[i]
print(sum)
```

35. MAX ELEMENT IN ARRAY

```
n = int(input("ENTER N : "))
arr = list(map(int,input().split()))
max = arr[0]
for i in range(1,n):
    if arr[i] > max:
        max = arr[i]
print(max)
```

36. MIN ELEMENT IN ARRAY

```
n = int(input("ENTER N : "))
arr = list(map(int,input().split()))
min = arr[0]
for i in range(1,n):
    if arr[i] < min:
        min = arr[i]
print(min)
```

37. FIND AVERAGE OF AN ARRAY

```
from math import sqrt
def average(n,arr):
    sum = 0
    for i in arr:
        sum = sum + i
    a = (sum)/n
    return a
```

38. SUM OF ALL PRIME NUMBERS IN AN ARRAY

```
from math import sqrt
def prime(n):
    if n < 2:
        return False
    i = 2
    end = int(sqrt(n))
    while(i <= end):
        if n%i == 0:
            return False
        i = i + 1
    return True
n = int(input("ENTER N : "))
arr = list(map(int,input().split()))
sum = 0
for i in arr:
    x = prime(i)
    if x == True:
        sum = sum + i
print(sum)
```

39. COUNT NUMBER OF PRIME NUMBERS IN AN ARRAY

```
def prime(n):
    if n < 2:
        return False
    i = 2
    end = int(sqrt(n))
    while(i <= end):
        if n%i == 0:
            return False
        i = i + 1
    return True
n = int(input("ENTER N : "))
arr = list(map(int,input().split()))
count = 0
for i in arr:
    x = prime(i)
    if x == True:
        count = count + 1
print(count)
```

40. LINEAR SEARCH

```
def liner(st,ch):
    n = len(st)
    for i in range(0,n):
        if st[i] == ch:
            return True
    return False
st = input("ENTER STRING : ")
ch = input("ENTER CH : ")
x = liner(st,ch)
print(x)
```

```
def liner(st,ch):
    n = len(st)
    for i in st:
        if i == ch:
            return True
    return False
st = input("ENTER STRING : ")
ch = input("ENTER CH : ")
x = liner(st,ch)
print(x)
```

41. BINARY SEARCH

```
def binarysearc(n,arr,item):
    start = 0
    end = n-1
    mid = (start+end)//2
    while(start <= end):
        if (arr[mid] == item):
            return True
        elif(item > arr[mid]):
            start = mid + 1
            mid = (start+end)//2
        else:
            end = mid - 1
            mid = (start+end)//2
    return False
n = int(input("ENTER THE LENGTH OF ARRAY : "))
arr = list(map(int,input("ENTER THE ARRAY : ").split()))
item = int(input("ENTER THE ITEM : "))
```

```
x = binarysearc(n,arr,item)
print(x)
```

42. REVERSE A STRING

```
def rever(s):
    s1 = ""
    i = len(s)-1
    while(i>=0):
        s1 = s1 + s[i]
        i = i - 1
    return s1
s = input()
print(rever(s))
```

```
def reverseee(s):
    s = list(s)
    start = 0
    end = len(s)-1
    while(start < end):
        s[start],s[end] = s[end],s[start]
        start = start + 1
        end = end - 1
    s = "".join(s)
    return s
s = input()
print(reverseee(s))
```

43. SELECTION SORT

```
def selectionsort(arr,n):
    for i in range(0,n):
        for j in range(i+1,n):
            if arr[i] > arr[j]:
                arr[i],arr[j]=arr[j],arr[i]
                # c = arr[i]
                # arr[i] = arr[j]
                # arr[j] = c+

                # arr[i] = arr[i] + arr[j]
                # arr[j] = arr[i] - arr[j]
                # arr[i] = arr[i] - arr[j]
    return arr
n = int(input("ENTER N : "))
```

```
arr = list(map(int,input().split()))
print(selectionsort(arr,n))
```

44. BUBBLE SORT

```
def bubble(n,arr):
    for i in range(0,n):
        swapped = 0
        for j in range(0,n-i-1):
            if arr[j] > arr[j+1]:
                arr[j],arr[j+1]=arr[j+1],arr[j]
                swapped = 1
        if swapped == 0:
            break
    return arr
n = int(input("ENTER N : "))
arr = list(map(int,input().split()))
print(bubble(n,arr))
```

45. SUM OF PAIRS == K

```
def pairsum(n,arr,k):
    count = 0
    for i in range(0,n):
        for j in range(i+1,n):
            if arr[i]+arr[j] == k:
                count = count + 1
    return count
n = int(input())
arr = list(map(int,input().split()))
k = int(input())
print(pairsum(n,arr,k))
```

46. SECOND HIGHEST ELEMENT

```
def secondHighest(n,arr):
    highest = float('-inf')
    second = float('-inf')

    for i in range(0,n):
        if arr[i] > highest:
            second = highest
            highest = arr[i]
        elif arr[i] > second and arr[i] != highest:
            second = arr[i]
    return second
```



```
n = int(input())
arr = list(map(int,input().split()))
print(secondHighest(n,arr))
```

47. ANNAGRAM STRING

```
def annagram(s1,s2):
    if sorted(s1) == sorted(s2):
        return True
    return False
s1 = input()
s2 = input()
print(annagram(s1,s2))
```

48. s = "I love you" : o/p = "you love I"

```
def fun(s):
    arr = s.split()
    arr = arr[::-1]
    s = " ".join(arr)
    return s
s = input()
x = fun(s)
print(x)
```

49. s = "I love you" : o/p = "I evol ouy"

```
def funcc(s):
    arr = s.split()
    s1 = " "
    for i in arr:
        s1 = s1 + i[::-1] + " "
    return s1
s = input()
print(funcc(s))

def func(s):
    arr = s.split()
    for i in arr:
        print(i[::-1],end=" ")

s = input()
func(s)
```

50. SPECIAL CHARECTERS IN STRING

```
def countspec(s):
    count = 0
    for i in s:
        if i.isalnum() == True or i.isspace() == True:
            pass
        else:
            count = count + 1
    return count
s = input()
x = countspec(s)
print(x)
```

51. sort an string which contains numbers and characters

```
s = "g45au96r7a54"
s1 = ""
s2 = ""
for i in s :
    if i.isdigit() == True :
        s1 = s1 + i
    else:
        s2 = s2 + i
s4 = sorted(s1)
s5 = sorted(s2)
s3 = s5 + s4
s3 = "".join(s3)
print(s3)
```

52. ASCII value sum

```
s = "gaurav"
sum = 0
for i in s:
    sum = sum + ord(i) # ordinate
print(sum)
```

53. s="abcd" : o/p = 1+2+3+4 i.e., 10

```
s = "abcd" # 1+2+3+4
sum = 0
for i in s:
    sum = sum + (ord(i)-96)
```

```
print(sum)
```

54. s = "abcd" : o/p = zyxw

```
s = "abcd"
s1 = ""
for i in s:
    s1 = s1 + chr(122-ord(i)+97)
print(s1)
```

55. replace a vowel with next vowel and consonent with next consonant

```
def replace(s):
    s = list(s)
    v = ['a','e','i','o','u']
    for i in range(len(s)):
        if s[i] in v:
            if s[i] == 'a':
                s[i] = 'e'
            elif s[i] == 'e':
                s[i] = 'i'
            elif s[i] == 'i':
                s[i] = 'o'
            elif s[i] == 'o':
                s[i] = 'u'
            elif s[i] == 'u':
                s[i] = 'a'
        else:
            if s[i] != 'z':
                s[i] = chr(ord(s[i])+1)
            else:
                s[i] = 'a'
    s = "".join(s)
    return s
s = input()
print(replace(s))
```

56. s = g3v2k5 o/p = gggvvkkkkk

```
def replace(s):
    n = len(s)
    s1 = ""
    for i in range(0,n-1,2):
        x = s[i]
        y = int(s[i+1])
        s1 = s1 + (x*y)
    return s1
```

```
s = input()
print(replace(s))
```

57. largest number in a string s = ga56ra96v123gk9654v2316k o/p : 9654

```
def fun(s):
    n = len(s)
    l = []
    s1 = ""
    for i in range(n):
        if s[i].isdigit() == True:
            s1 = s1 + s[i]
        if s[i].isalpha() == True:
            if s1 != "":
                l.append(s1)
                s1 = ""
        if i == n-1 :
            if s1 != "":
                l.append(s1)

    l2 = []
    for i in l:
        l2.append(int(i))
    return max(l2)
s = input()
print(fun(s))
```

```
def fun(s):
    n = len(s)
    l = []
    s1 = ""
    for i in range(n):
        if s[i].isdigit() == True:
            s1 = s1 + s[i]
        if s[i].isalpha() == True:
            if s1 != "":
                l.append(s1)
                s1 = ""
        if i == n-1 :
            if s1 != "":
                l.append(s1)

    m = len(l)
    l2 = [0] * m # [0,0,0,0]
    for i in range(m):
        l2[i] = int(l[i])
```

```
    return max(l2)
s = input()
print(fun(s))
```

58. count vowels in an array

```
s = input()
v = ['a','e','i','o','u','A','E','I','O','U']
count = 0
for i in s:
    if i in v:
        count = count + 1
print(count)
```

59. sum of even and odd numbers in an array and print their absolute difference

```
arr = [10,20,5,6,7,21]
even = 0
odd = 0
for i in arr:
    if i%2 == 0:
        even = even + i
    else:
        odd = odd + i
print(abs(odd-even))
```

60. remove duplicates from a sorted array

```
def fun(n,arr):
    if n == 1:
        return n
    inx = 1
    for i in range(1,n):
        if arr[i] != arr[i-1]:
            arr[inx] = arr[i]
            inx = inx + 1
    return inx
n = int(input())
arr = list(map(int,input().split()))
x = fun(n,arr)
for i in range(0,x):
    print(arr[i],end = " ")
```

61. print all substrings

```
def printsubstring(s):
    n = len(s)
    for i in range(0,n):
        s1 = ""
        for j in range(i,n):
            s1 = s1 + s[j]
            print(s1)
s = input()
printsubstring(s)
```

62. REMOVE DUPLICATES FROM SORTED AND UNSORTED ARRAY

```
def duplicates(arr):
    d = {}
    for i in arr:
        d[i] = 0
    for k in d.keys():
        print(k)
arr = list(map(int,input().split()))
duplicates(arr)
```

63. REMOVE DUPLICATES FROM STRING

```
def remove_duplicates(s):
    d = {}
    for i in s:
        d[i] = 0
    s1 = ""
    for k in d.keys():
        s1 = s1 + k
    return s1
s = input()
print(remove_duplicates(s))
```

64. COUNTING FREQUENCY OF EACH CHARACTER IN STRING

```
def freq_count(s):
    d = {}
    for i in s:
        d[i] = 0
    for i in s:
        d[i] = d[i] + 1
    for k,v in d.items():
        print(k," occurred ",v," times")
```

```
s = input()
freq_count(s)
```

65. REMOVE DUPLICATES

```
def remove_duplicates_3(s):
    d = {}
    for i in s:
        d[i] = 0
    for i in s:
        d[i] = d[i] + 1
    for k,v in d.items():
        if v == 1:
            print(k, end= "")
s = input()
remove_duplicates_3(s)
```

66. FIND DUPLICATES

```
def find_duplicates_arr(n,arr):
    d={}
    for i in arr:
        d[i] = 0
    for i in arr:
        d[i] = d[i] + 1
    for k,v in d.items():
        if v > 1:
            print(k,end=" ")
n = int(input())
arr = list(map(int,input().split()))
find_duplicates_arr(n,arr)
```

67. PRINT OCCURANCE OF EACH ELEMENT IN DECENDING ORDER

```
arr = list(map(int,input().split()))
d = {}
for i in arr:
    d[i] = 0
for i in arr:
    d[i] = d[i] + 1
d = sorted(d.items(),key = lambda x : x[1]) # x[1] means values and x[0] means keys
d = dict(d)                                # so we want to sort according to keys
then write x[0]
for k,v in d.items():
    print(k , "OCCURED", v)
```

68. PRINT OCCURANCE OF EACH ELEMENT IN ASCENDING ORDER

```
arr = list(map(int,input().split()))
d = {}
for i in arr:
    d[i] = 0
for i in arr:
    d[i] = d[i] + 1
d = sorted(d.items(),key = lambda x : x[1],reverse=True)
d = dict(d)
for k,v in d.items():
    print(k , "OCCURED", v)
```

69. MAX LENGTH OF A SUBARRAY WHOSE SUM == 0

```
arr = [15, -2, 2, -8, 1, 7, 10, 23]
n = len(arr)-1
presum = {}
sum = 0
max_len = 0
for i in range(0,n):
    sum = sum + arr[i]
    if sum == 0:
        max_len = i + 1
    if sum in presum:
        max_len = max(max_len,i - presum[sum])
    else:
        presum[sum] = i
print(max_len)
```

70. COUNT NUMBER OF PAIRS WHOSE SUM == K

```
arr = [1,5,7,-1,5]
k = 6
count = 0
d = {}
for i in arr:
    complement = k - i
    if complement in d:
        count = count + d[complement]
    d[i] = d.get(i,0) + 1
print(count)
```


71. GIVEN THE ARRAY IS SORTED FIND IF THERE EXISTS A PAIR WITH SUM == k

```
def main(arr,k):
    st = 0
    end = len(arr) - 1
    while st < end:
        sum = arr[st] + arr[end]
        if sum == k:
            return True
        elif sum > k:
            end = end - 1
        else:
            st = st + 1
    return False
n = int(input())
arr = list(map(int,input().split()))
k = int(input())
print(main(arr,k))
```

72. GIVEN THE ARRAY IS SORTED FIND IF THERE EXISTS A PAIR WITH PRODUCT == k

```
def main(arr,k):
    st = 0
    end = len(arr) - 1
    while st < end:
        sum = arr[st] * arr[end]
        if sum == k:
            return (arr[st],arr[end])
        elif sum > k:
            end = end - 1
        else:
            st = st + 1
    return False
n = int(input())
arr = list(map(int,input().split()))
k = int(input())
a = main(arr,k)
print(a)
```

73. ROTATE AN ARRAY BY K

```
def rotateby_k(arr,st,end):
    while(st < end):
        arr[st],arr[end]=arr[end],arr[st]
        st = st + 1
        end = end - 1
    return arr
n = int(input())
arr = list(map(int,input().split()))
k = int(input())
if k > n:
    k = k % n
rotateby_k(arr,0,n-1)
rotateby_k(arr,0,k-1)
rotateby_k(arr,k,n-1)
print(arr)
```

74. FACTORIAL USING RECURSSION

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)
n = int(input())
print(factorial(n))
```

75. FIBONACCI SERIES USING RECURSSION

```
def n_th_fibo(n):
    if n <= 1:
        return n
    return n_th_fibo(n-1) + n_th_fibo(n-2)
n = 10
print(n_th_fibo(n))

def n_th_fibo(n):
    if n <= 1:
        return n
    return n_th_fibo(n-1) + n_th_fibo(n-2)
n = 10
for i in range(0,n):
    x = n_th_fibo(i)
    print(x, end= " ")
```

76. GCD USING RECURSION OR HCF

```
def hcf(a,b):  
    if b == 0:  
        return a  
    return hcf(b,a%b)  
a = int(input())  
b = int(input())  
x = hcf(a,b)  
print(x)
```

77. TOWER OF HANOI WITH 4 VARIABLES

78. IF A NUMBER IN FIBO USING RECURSSION